NOVANECTAR SERVICES PVT. LTD.

INTERNSHIP REPORT

Data Analytics Internship

Exploratory Data Analysis of Swiggy Dataset

This internship report is submitted in partial fulfilment for the completion of the internship program

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Acknowledgement

I would like to express my sincere gratitude to everyone who contributed to the successful completion of this project titled “Automated Exploratory Data Analysis (EDA) using Streamlit for Swiggy Dataset Analysis.”

First and foremost, I would like to thank my mentors and instructors for their continuous guidance, encouragement, and valuable insights throughout the development of this project. Their support helped me gain a deeper understanding of data analysis concepts, Python programming, and interactive application development.

I am also grateful to the open-source community for providing powerful libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Streamlit, which made it possible to build an efficient and interactive EDA application. These tools played a crucial role in performing data cleaning, visualization, and generating AI-style insights through rule-based logic.

Special thanks to the data providers whose Swiggy dataset enabled practical, real-world analysis and helped me understand food delivery trends, customer behavior, and data quality challenges.

INTRODUCTION

In today's data-driven world, the ability to quickly explore, understand, and derive insights from large datasets is essential for effective decision-making. Exploratory Data Analysis (EDA) plays a crucial role in identifying patterns, trends, anomalies, and data quality issues before applying advanced analytical or machine learning techniques. However, traditional EDA often requires extensive manual coding and time-consuming analysis.

This project, titled "Automated Exploratory Data Analysis (EDA) using Streamlit for Swiggy Dataset Analysis," focuses on developing an interactive and user-friendly web application that simplifies the EDA process. Built using Streamlit, the application enables users to perform data exploration, visualization, and statistical analysis with minimal coding effort. The system leverages popular Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn to efficiently process and visualize the data.

The application simulates AI-style insights using rule-based logic to automatically highlight important patterns, trends, missing values, and potential analytical opportunities within the Swiggy food delivery dataset. This approach helps analysts and beginners alike to gain meaningful insights quickly while improving data understanding and quality assessment.

By applying automated EDA techniques to a real-world Swiggy dataset, this project demonstrates how interactive dashboards can enhance data analysis efficiency, reduce manual effort, and support informed decision-making in the food delivery domain.

Swiggy Generative AI EDA – Project Report

1. Project Overview

This project presents an **Automated Exploratory Data Analysis (EDA) application** built using **Streamlit** for analysing a Swiggy food delivery dataset. The application is designed to provide a fast, interactive, and visually intuitive way to understand large datasets without requiring extensive manual coding. It simulates **AI-style insights** using rule-based logic to assist analysts in identifying patterns, data quality issues, and analytical opportunities.

2. Objectives

The primary objectives of this project are:

- To perform automated exploratory data analysis on the Swiggy dataset
- To visualize key dataset characteristics interactively
- To identify missing values, distributions, and categorical dominance
- To generate rule-based, AI-like insights for decision-making support
- To demonstrate how Streamlit can be used to build data-driven web applications

3. Tools and Technologies Used

- **Python** – Core programming language
- **Streamlit** – Web application framework for data apps
- **Pandas** – Data manipulation and analysis
- **Matplotlib** – Basic data visualization
- **Seaborn** – Statistical data visualization

Environment: **PyCharm**

Execution Platform**: Localhost (Web Browser)**

4. Methodology

The project was carried out using the following steps:

1. **Data Collection:** Loading the Swiggy dataset using Pandas
2. **Data Cleaning:** Handling missing values and duplicates
3. **Exploratory Data Analysis:** Identifying patterns and trends
4. **Visualization:** Creating graphs and charts
5. **Dashboard Development:** Using Streamlit for interactivity
6. **Deployment:** Running the application on localhost
7. **7. Functions and Techniques Used**

5.1 Streamlit Functions

- st.set_page_config() – Page configuration
- st.title(), st.subheader(), st.write() – Display content
- st.sidebar.selectbox() – User input for filtering
- st.dataframe() – Display dataset
- st.pyplot() – Display visualizations
- @st.cache_data – Improve performance

5.2 Pandas Functions

pd.read_csv()

- Used to load the Swiggy dataset from a CSV file into a DataFrame.
- **df.head()**
- Used to display the first five rows of the dataset for preview.

- **df.shape**
- Used to get the total number of rows and columns in the dataset.
- **df.isnull()**
- Used to identify missing values in the dataset.
- **df.isnull().sum()**
- Used to calculate missing values for each column.
- **df.isnull().sum().sum()**
- Used to calculate the total number of missing values in the dataset.
- **df.dtypes**
- Used to identify the data type of each column.
- **df.describe()**
- Used to generate statistical summary of numerical columns.
- **df.select_dtypes()**
- Used to select numerical and categorical columns separately.
- **df.columns**
- Used to retrieve all column names from the DataFrame.
- **df[column].dtype**
- Used to check the data type of a selected column.
- **df[column].value_counts()**
- Used to calculate frequency of values in categorical columns

6. Dataset Description

The Swiggy dataset contains information related to restaurants and food delivery services. It includes both:

- **Numerical features** such as prices, ratings, or delivery-related metrics
- **Categorical features** such as restaurant names, cuisines, locations, and food categories

The dataset enables analysis of customer preferences, pricing patterns, restaurant segmentation, and data completeness.

7. Application Architecture and Workflow

1 Page Configuration

The application begins by configuring the Streamlit page with a custom title and wide layout to enhance readability and visualization space.

2 Data Loading and Caching

- The dataset is loaded from a CSV file using Pandas.
- Streamlit's @st.cache_data decorator is used to cache the dataset, improving performance by avoiding repeated file reads during reruns.

3 Dataset Preview

- The first few rows of the dataset are displayed using an interactive table.
- This allows users to quickly understand the structure and content of the data.

4 Dataset Overview Metrics

Key metrics are displayed using Streamlit's st.metric feature:

- Total number of rows
- Total number of columns
- Total count of missing values

These metrics provide an immediate snapshot of dataset size and quality.

8. Exploratory Data Analysis

The EDA focuses on:

- City-wise distribution of restaurants
- Popular cuisines offered on Swiggy
- Relationship between ratings and price
- Top-rated restaurants
- Price range analysis across cities

9. Data Visualization

Various visualizations were used:

- Bar charts for city and cuisine analysis
- Histograms for rating distribution
- Box plots for price vs rating analysis

These visualizations help in understanding patterns and trends effectively.

Running the Streamlit Application

The Streamlit application was executed using the following commands:

```
pip install streamlit

streamlit run app.py

python -m streamlit run app.py
```

The application runs on:

http://localhost:8501

Challenges Faced

- Handling missing and inconsistent data
- Designing an effective Streamlit layout
- Optimizing performance for large datasets
- Implementing dynamic filters

10. AI-Generated (Rule-Based) Insights

The application includes a rule-based insight engine that simulates AI-driven observations:

- Identifies whether the dataset is large enough for trend analysis
- Flags the presence of missing values
- Detects analytical potential based on the number of numerical features
- Highlights segmentation opportunities based on categorical features
- These insights provide high-level guidance without requiring manual interpretation.

11. Results and Key Findings

- The dataset structure and size can be quickly understood
- Data quality issues such as missing values are easily detected
- Pricing, rating, and categorical dominance patterns can be visually analyzed
- Automated insights reduce the effort required for initial data understanding

12. Conclusion

The Swiggy Generative AI EDA application successfully demonstrates how **Streamlit** can be used to automate exploratory data analysis and deliver meaningful insights through an interactive dashboard. By combining visual analytics with rule-based AI-style insights, the app enhances productivity and accessibility for data analysts and beginners alike.

This project serves as a strong foundation for advanced enhancements such as:

- True generative AI insights using LLMs
- Predictive modelling and recommendation systems
- Advanced filtering and comparative analysis

13. Future Scope

- Integration of real AI/LLM-based narrative insights
- Addition of correlation heatmaps and geospatial analysis
- Deployment on cloud platforms for public access
- Support for uploading custom datasets dynamically
- **User-Driven Data Uploads**: Allow users to upload their own datasets dynamically, making the application reusable beyond the Swiggy dataset.

14. Project Snapshots

This section presents visual snapshots of the Streamlit-based Swiggy Generative AI EDA application. These snapshots help illustrate the functionality, layout, and analytical capabilities of the dashboard.

14.1 Home Dashboard and Dataset Preview

Figure 1: Home page showing the application title and dataset preview.

🍽️ Swiggy Dataset – Generative AI EDA

Automated exploratory data analysis with AI-style insights

Dataset Preview

	ID	Area	City	Restaurant	Price	Avg ratings	Total ratings	Food type	Address	Delivery time
0	211	Koramangala	Bangalore	Tandoor Hut	300	4.4	100	Biryani,Chinese,North Indian,South Indian	5Th Block	59
1	221	Koramangala	Bangalore	Tunday Kababi	300	4.1	100	Mughlai,Lucknowi	5Th Block	56
2	246	Jogupalya	Bangalore	Kim Lee	650	4.4	100	Chinese	Double Road	50
3	248	Indiranagar	Bangalore	New Punjabi Hotel	250	3.9	500	North Indian,Punjabi,Tandoor,Chinese	80 Feet Road	57
4	249	Indiranagar	Bangalore	Nh8	350	4	50	Rajasthani,Gujarati,North Indian,Snacks,Desserts,Beverages,Thalis,Chaat	80 Feet Road	63

14.2 Dataset Overview Metrics

Figure 2: Overview of rows, columns, and missing values.

Dataset Overview

Rows	Columns	Missing Values
8680	10	0

14.3 Missing Values Analysis

Figure 3: Bar chart representing missing values across columns.

Statistical Summary

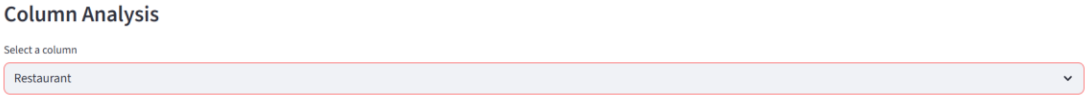
	ID	Price	Avg ratings	Total ratings	Delivery time
count	8680	8680	8680	8680	8680
mean	244812.0714	348.4445	3.6551	156.6348	53.9671
std	158671.6172	230.9401	0.6476	391.448	14.2923
min	211	0	2	20	20
25%	72664	200	2.9	50	44
50%	283442	300	3.9	80	53
75%	393425.25	400	4.2	100	64
max	466928	2500	5	10000	109

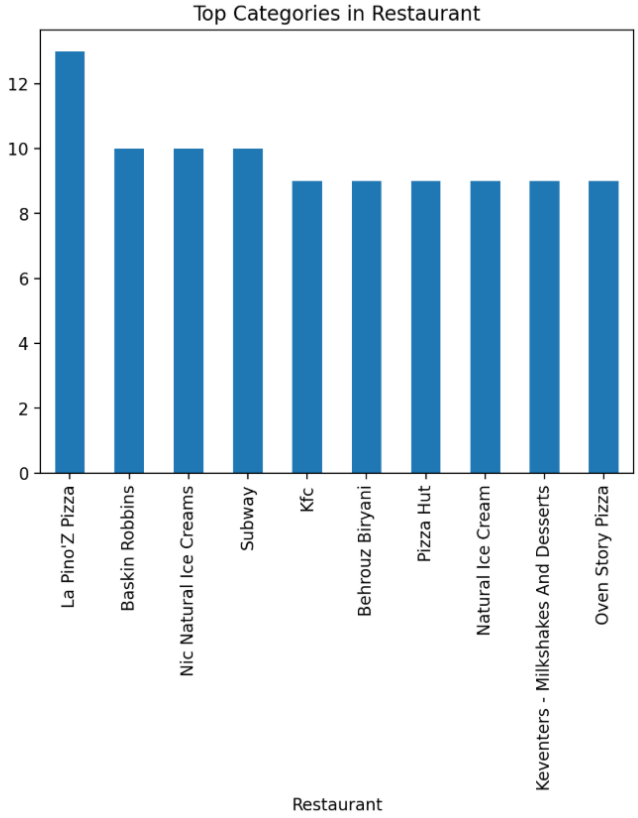
Missing Values Analysis

No missing values found 🎉

14.4 Column Analysis Visualization

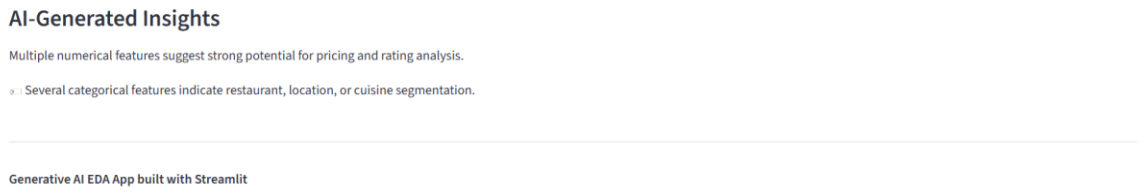
Figure 4: Distribution and categorical analysis of selected columns.





14.5 AI-Generated Insights Section

Figure 5: Rule-based AI-generated insights for the Swiggy dataset.



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